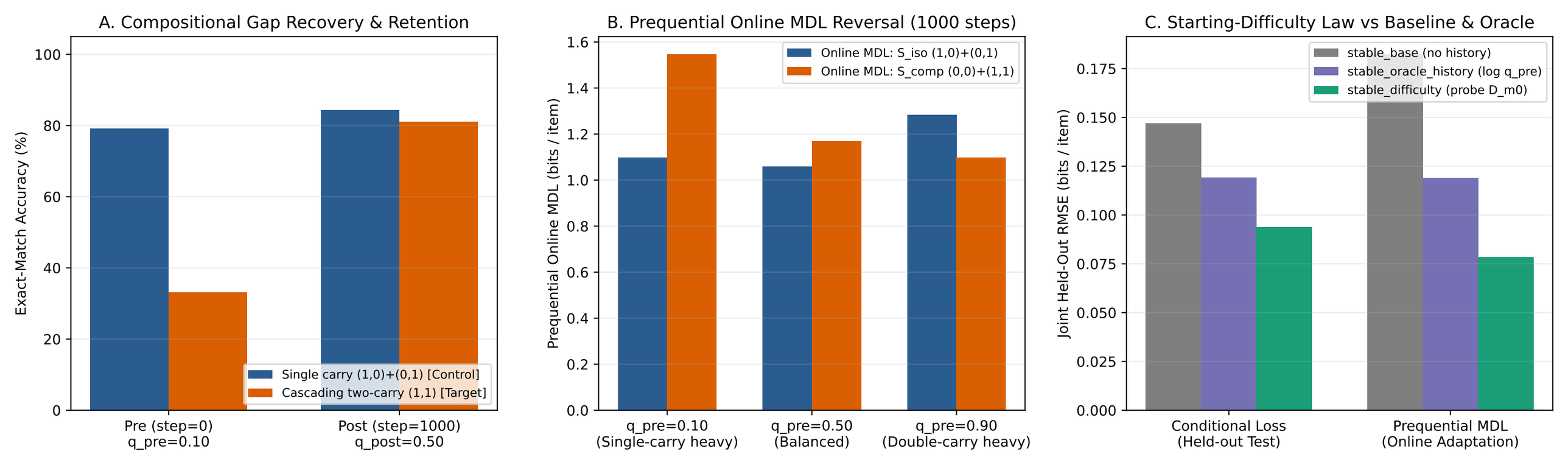
A. Compositional Gap Recovery & Retention
Exact-Match Accuracy (%)
100
80
60
40
20
0
Single carry (1,0)+(0,1) [Control]
Cascading two-carry (1,1) [Target]
Pre (step=0)
q_pre=0.10
Post (step=1000)
q_post=0.50
B. Prequential Online MDL Reversal (1000 steps)
Prequential Online MDL (bits / item)
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Online MDL: S_iso (1,0)+(0,1)
Online MDL: S_comp (0,0)+(1,1)
q_pre=0.10
(Single-carry heavy)
q_pre=0.50
(Balanced)
q_pre=0.90
(Double-carry heavy)
C. Starting-Difficulty Law vs Baseline & Oracle
Joint Held-Out RMSE (bits / item)
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
stable_base (no history)
stable_oracle_history (log q_pre)
stable_difficulty (probe D_m0)
Conditional Loss
(Held-out Test)
Prequential MDL
(Online Adaptation)